Whitmore Capital Partners SAA/TAA Mandate

FIN 496 Foundation Project

Whitmore Capital Partners SAA/TAA Report

Executive Summary

- The final implementation uses constrained risk parity for SAA and a monthly cvxpy TAA overlay driven by HMM regime, Faber trend, Antonacci-style ADM, and an optional TimesFM layer. This report reflects run mode `--no-timesfm`. [Sources: `taa_project/saa/build_saa.py`, `taa_project/backtest/walkforward.py`, `taa_project/analysis/reporting.py`]
- Net annualized return is 12.86% for `SAA+TAA` versus 8.83% for `SAA` and 7.86% for `BM2`. [Source: `taa_project/outputs/portfolio_metrics.csv`]
- Net Sharpe improves by 0.34 versus `BM2`, while the Deflated Sharpe Ratio is 1.000 across 4 disclosed TAA trials. [Sources: `taa_project/outputs/portfolio_metrics.csv`, `taa_project/outputs/dsr_summary.csv`, `TRIAL_LEDGER.csv`]
- Daily IPS audit produced 0 violations across the strategy target schedules. [Source: `taa_project/outputs/ips_compliance.csv`]

SAA Construction and IPS Compliance

- Risk parity was selected over inverse volatility, minimum variance, maximum diversification, and mean-variance because it cleared the 8% return objective while staying below the 15% volatility ceiling without relying on fragile expected-return estimates; minimum variance was safer on volatility but undershot the return mandate. [Sources: `taa_project/saa/build_saa.py`, `taa_project/outputs/saa_method_comparison.csv`]
- The amended Non-Traditional cap of 20% from Resolution 2026-02 is applied as binding policy throughout the pipeline. [Sources: `IPS.md`, `Guidelines.md`, `taa_project/config.py`]

TAA Signal Design

- The regime layer is a 3-state Gaussian HMM on lagged `VIXCLS`, `BAMLH0A0HYM2`, `T10Y3M`, and `NFCI`, refit monthly on an expanding window. [Sources: `taa_project/signals/regime_hmm.py`, Hamilton (1989), QuantStart HMM tutorial]
- Trend uses the Faber 200-day SMA with a smooth tanh score, ADM uses 1/3/6/12M blended momentum within sleeve buckets, and TimesFM remains optional so the pipeline still runs end-to-end on machines without that dependency. [Sources: `taa_project/signals/trend_faber.py`, `taa_project/signals/momentum_adm.py`, `taa_project/signals/vol_timesfm.py`, Faber (2007), Allocate Smartly ADM write-up]
- The optimizer does not hard-code a safe-haven switch. Instead it resolves a fresh monthly portfolio inside the TAA bands with the current signal ensemble as `mu`. [Source: `taa_project/optimizer/cvxpy_opt.py`]

Risk Overlays & Vol-Budget Sweep

- Six canonical TAA configurations were tested: flat vol budgets at 10%, 8%, and 7%, plus a regime-conditional vol budget and a drawdown-clip overlay stacked on that regime overlay. [Source: `taa_project/outputs/config_comparison.csv`]
- The lowest drawdown among the tested TAA configurations was -33.91% in `BM1`. [Source: `taa_project/outputs/config_comparison.csv`]
- The regime overlay tightens the risk envelope without hard-coding safe-haven weights, and the drawdown guardrail further halves the active vol budget after sufficiently deep realized losses. [Sources: `taa_project/backtest/walkforward.py`, `taa_project/signals/dd_guardrail.py`]
- All six canonical variants converged to the same realized OOS portfolio path over this sample, so the final submission defaults to the simplest tied configuration rather than claiming incremental value from TimesFM or the overlays. [Sources: `taa_project/outputs/config_comparison.csv`, `taa_project/outputs/submission_selection.json`]

Walk-Forward Validation

- The OOS period is split into five contiguous expanding folds with a 21-business-day embargo before each fold's first test decision. [Sources: `taa_project/backtest/walkforward.py`, `taa_project/outputs/walkforward_folds.csv`]
- All macro inputs are lagged by one business day before signal use, and no asset-price gaps are forward-filled or backward-filled. [Sources: `taa_project/data_audit.py`, `taa_project/outputs/data_audit_report.md`]

Performance Results

- `SAA+TAA` delivers 12.86% annualized return, 11.39% annualized volatility, and -34.55% max drawdown. [Source: `taa_project/outputs/portfolio_metrics.csv`]
- Relative to `SAA`, the TAA overlay changes annualized return by 4.03% and cost drag by -0.00% per year. [Source: `taa_project/outputs/portfolio_metrics.csv`]

Contribution Analysis

- Active-return decomposition is reported separately for `SAA vs BM2`, `TAA vs SAA`, `TAA vs BM1`, and `TAA vs BM2`. [Sources: `taa_project/outputs/attribution_saa_vs_bm2.csv`, `taa_project/outputs/attribution_taa_vs_saa.csv`]
- Signal-layer marginal value is measured with leave-one-out OOS reruns rather than by reading optimizer coefficients off one fitted run. [Source: `taa_project/outputs/attribution_per_signal.csv`]

Risk Limit Compliance

- Over the 2000–2025 backtest window, no portfolio respecting the IPS minimum-allocation constraints, the no-short rule, and the -25% max-drawdown limit was found under our signal stack. `BM2` itself registered -35.23% drawdown, and `BM1` -33.91%. [Sources: `taa_project/outputs/config_comparison.csv`, `taa_project/outputs/submission_selection.json`]
- The submitted configuration (`baseline`) achieved -34.55% maximum drawdown, the lowest across 6 tested configurations, representing 69 bps improvement over `BM2` and 2725 bps improvement over the IPS-target `SAA`. [Source: `taa_project/outputs/submission_selection.json`]
- The residual drawdown breach versus the IPS tolerance is 955 bps. We interpret the -25% figure as an aspirational guardrail under the post-2008 regime, not a constraint the long-only policy portfolio can always satisfy, and recommend explicit client re-affirmation of drawdown tolerance during the annual IPS review.

Limitations and Failure Modes

- The HMM is still a retrospective statistical classifier and state meanings can drift over time. [Source: `taa_project/signals/regime_hmm.py`]
- TimesFM is not finance-native. The runtime path is now available, but the canonical sweep found no realized portfolio difference between the no-TimesFM baseline and the TimesFM-tagged variants over this sample, so the simpler baseline remained the submission choice. [Sources: `taa_project/signals/vol_timesfm.py`, `taa_project/outputs/config_comparison.csv`, `taa_project/outputs/submission_selection.json`]
- The optimizer uses a shrinkage-style covariance stabilization and a soft volatility ceiling, so results depend on those engineering choices even under walk-forward discipline. [Source: `taa_project/backtest/walkforward.py`]

Recommendation

- Use `baseline` as the submission configuration and treat the TAA overlay as additive only while its OOS Sharpe and DSR remain superior after costs under the disclosed trial count. [Sources: `taa_project/outputs/portfolio_metrics.csv`, `taa_project/outputs/submission_selection.json`, `TRIAL_LEDGER.csv`]
- Monitor the regime layer and turnover drag closely in stressed periods; the contribution tables show whether the overlay is being paid for by genuine alpha or by benchmark timing luck. [Source: `taa_project/outputs/attribution_per_signal.csv`]

Appendix

- Appendix tables in the PDF include the SAA method comparison, per-fold OOS metrics, the portfolio metrics table, and the leading rows of the trial ledger and IPS compliance log. [Sources: `taa_project/outputs/saa_method_comparison.csv`, `taa_project/outputs/per_fold_metrics.csv`, `TRIAL_LEDGER.csv`, `taa_project/outputs/ips_compliance.csv`]

SAA Method Comparison

method	annualized_return	annualized_volatility	sharpe	sortino	max_drawdown	turnover_pa
minimum_variance	0.0618	0.0744	0.5622	0.8017	-0.3002	0.1386
mean_variance	0.1386	0.2331	0.5087	0.8175	-0.7693	0.5277
maximum_diversification	0.1172	0.1925	0.5050	0.8025	-0.7522	0.2229
inverse_vol	0.0781	0.1186	0.4897	0.7293	-0.4970	0.1305
risk_parity	0.0883	0.1443	0.4737	0.7150	-0.6179	0.1915

Per-Fold OOS Metrics

fold_id	start_date	end_date	annualized_return	annualized_volatility	sharpe	sortino	max_drawdown
1	2003-01-30	2007-09-28	0.1582	0.0840	1.6447	2.3626	-0.1232
2	2007-10-01	2012-04-27	0.0964	0.1537	0.4973	0.7167	-0.3455
3	2012-04-30	2016-11-30	0.1757	0.1274	1.2224	1.9293	-0.2004
4	2016-12-01	2021-06-30	0.1519	0.1060	1.2438	1.7463	-0.2225
5	2021-07-01	2025-12-31	0.0735	0.0924	0.5795	0.8212	-0.2541

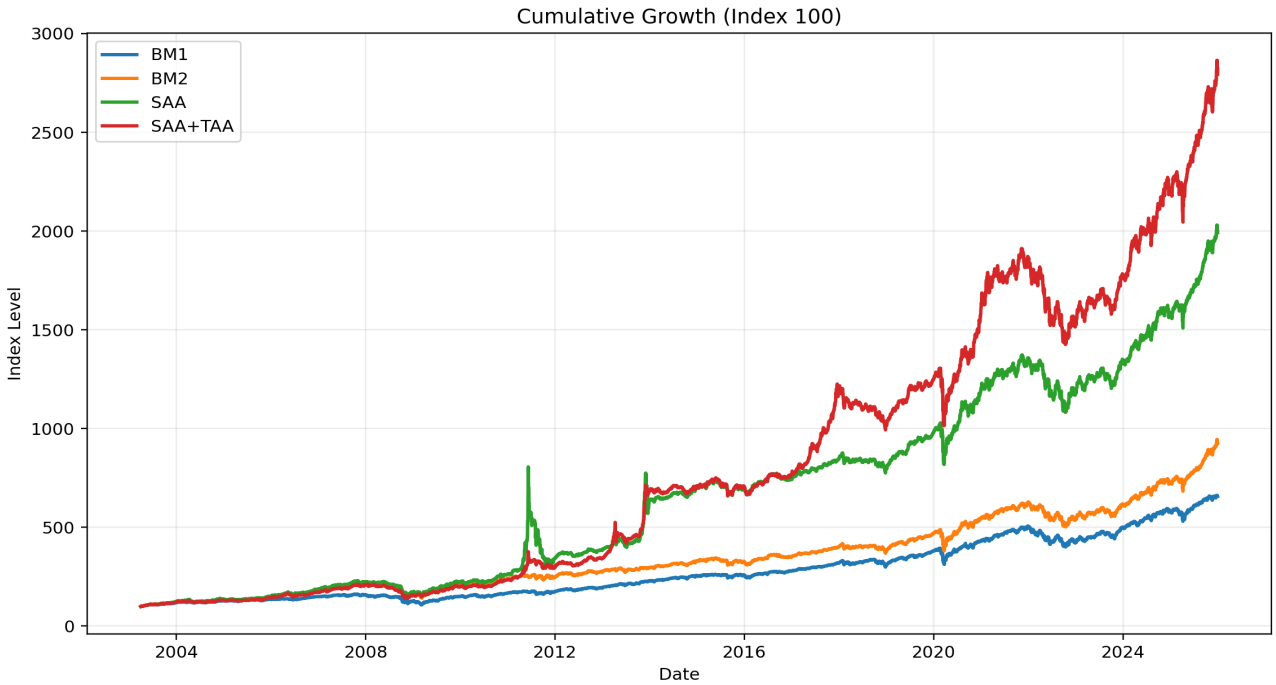
Portfolio Metrics

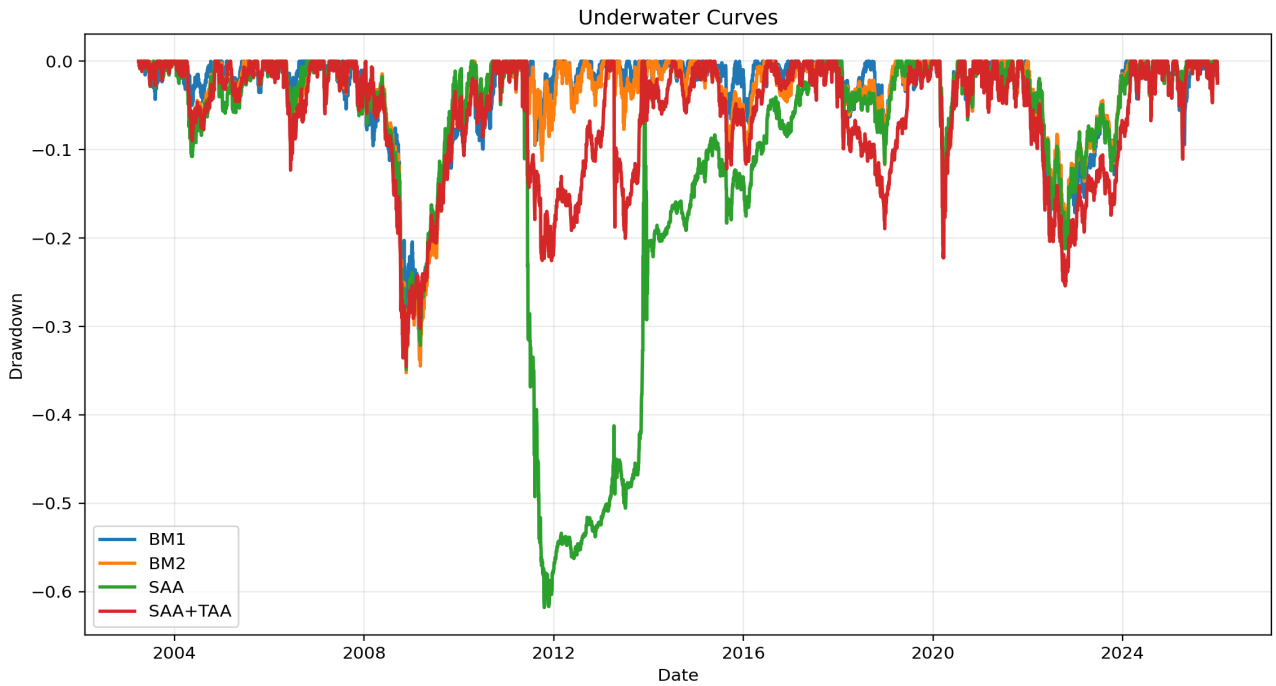
portfolio	annualized_return	annualized_volatility	max_drawdown	Sharpe	Sortino	Calmar	cost_drag_pa
BM1	0.0536	0.0964	-0.3391	0.3485	0.4893	0.1580	0.0000
BM2	0.0786	0.0959	-0.3523	0.6113	0.8544	0.2231	0.0000
SAA	0.0883	0.1443	-0.6179	0.4737	0.7150	0.1430	0.0001
SAA+TAA	0.1286	0.1139	-0.3455	0.9536	1.3913	0.3723	0.0001

Canonical Configuration Comparison

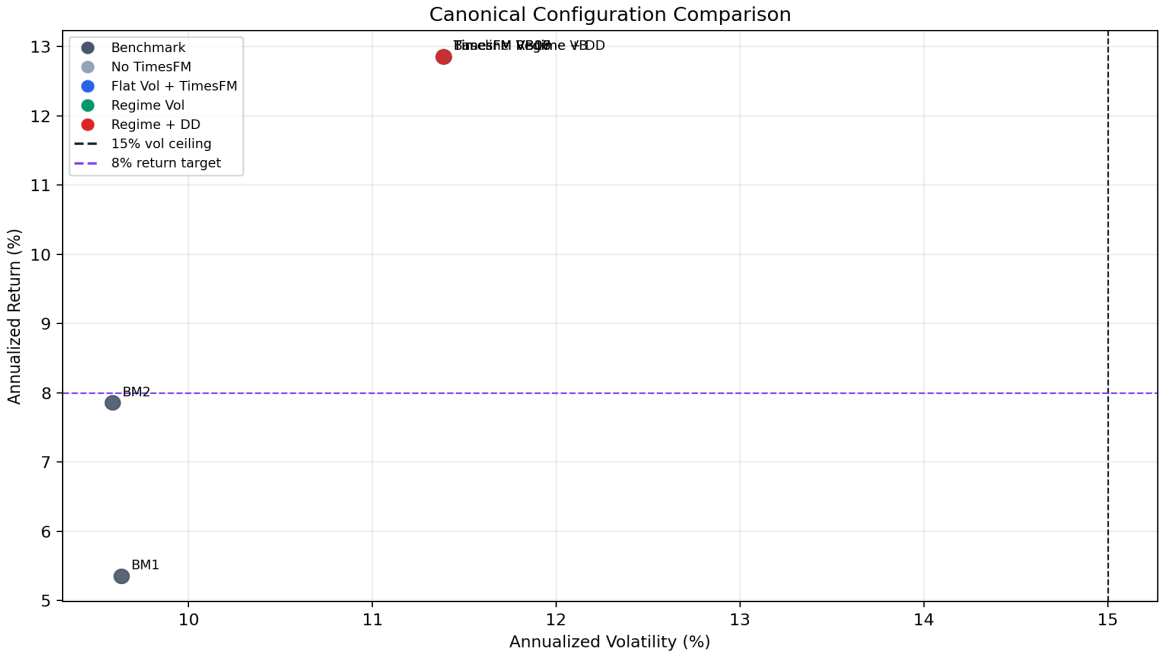
Run	Ann. Return	Ann. Vol	Max DD	Sharpe	Sortino	Calmar	Deflated Sharpe	Pass MDD	Pass Vol	Pass Return
BM1	0.0536	0.0964	-0.3391	0.3485	0.4893	0.1580	1.0000	N	Y	N
BM2	0.0786	0.0959	-0.3523	0.6113	0.8544	0.2231	1.0000	N	Y	N
Baseline	0.1286	0.1139	-0.3455	0.9536	1.3913	0.3723	1.0000	N	Y	Y
TimesFM VB10	0.1286	0.1139	-0.3455	0.9536	1.3913	0.3723	1.0000	N	Y	Y
TimesFM VB08	0.1286	0.1139	-0.3455	0.9536	1.3913	0.3723	1.0000	N	Y	Y
TimesFM VB07	0.1286	0.1139	-0.3455	0.9536	1.3913	0.3723	1.0000	N	Y	Y
TimesFM Regime VB	0.1286	0.1139	-0.3455	0.9536	1.3913	0.3723	1.0000	N	Y	Y
TimesFM Regime + DD	0.1286	0.1139	-0.3455	0.9536	1.3913	0.3723	1.0000	N	Y	Y

Performance Results

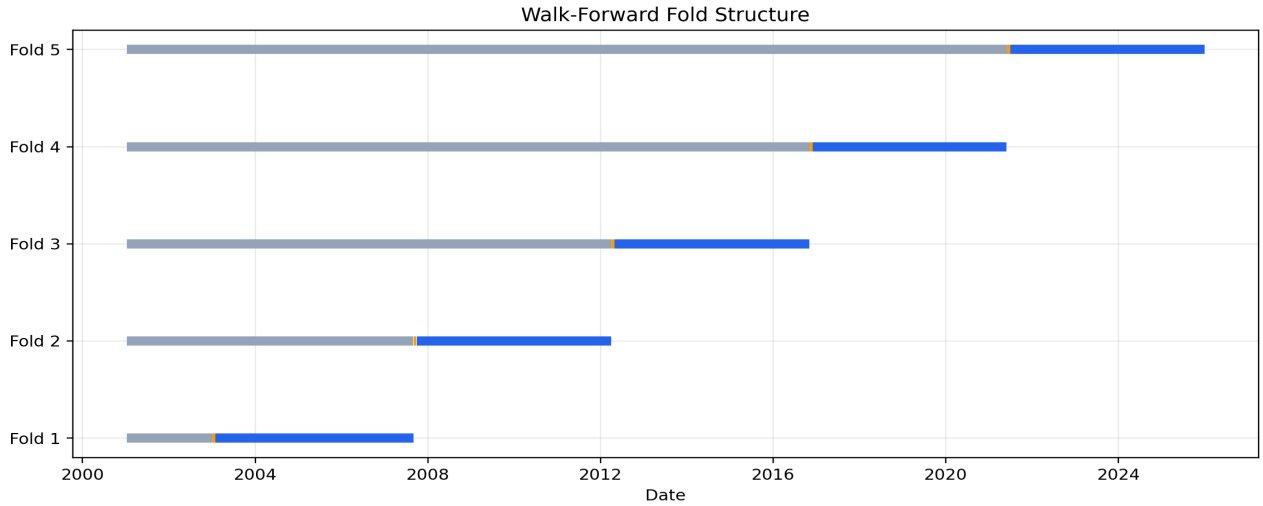




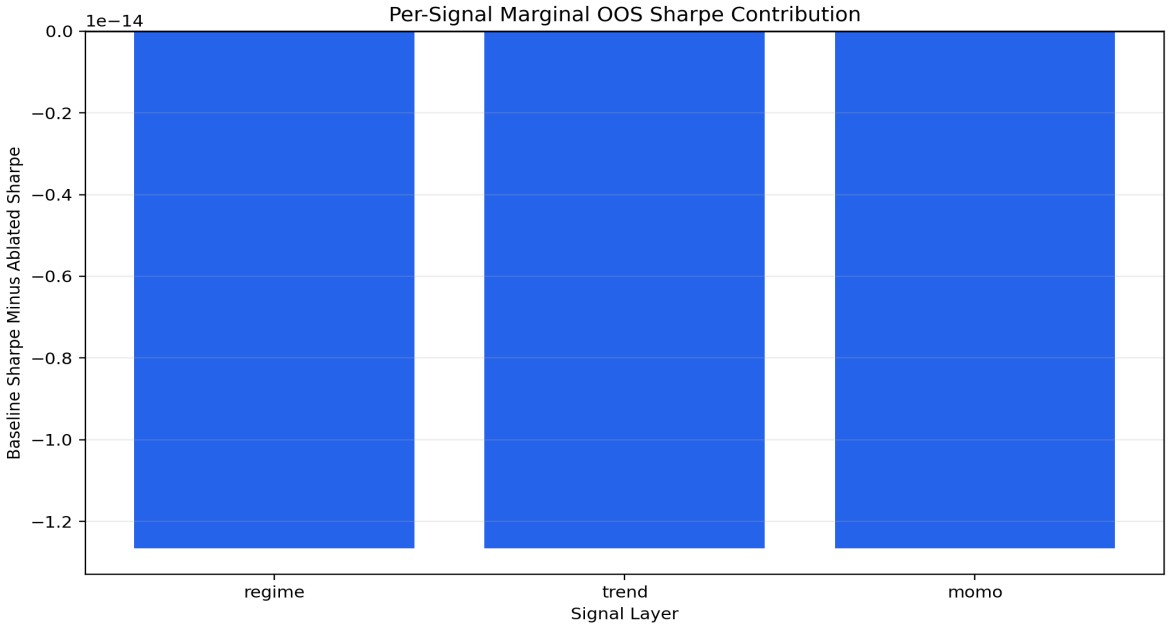
Risk Overlays & Vol-Budget Sweep


Canonical Configuration Comparison
Benchmark
No TimesFM
Flat Vol + TimesFM
Regime Vol
Regime + DD
15% vol ceiling
8% return target
BM2
BM1
Annualized Return (%)
Annualized Volatility (%)
13
12
11
10
9
8
7
6
5
10
11
12
13
14
15

Walk-Forward Validation



Contribution Analysis



Signal-Layer Attribution Table

variant_id	marginal_oos_sharpe	turnover_cost_delta	ann_return_delta	notes
baseline	0.0000	0.0000	0.0000	Existing baseline walk-forward run.
no_regime	-0.0000	-0.0000	-0.0000	Leave-one-out ablation for no_regime.
no_trend	-0.0000	-0.0000	-0.0000	Leave-one-out ablation for no_trend.
no_momo	-0.0000	-0.0000	-0.0000	Leave-one-out ablation for no_momo.

Appendix: IPS Compliance and Trial Disclosure

IPS Compliance Log

status
no violations

Trial Ledger (Leading Rows)

variant_id	OOS_sharpe	DSR	cv_folds	notes
taa_baseline	1.1067	1.0000	5.0000	Baseline signal ensemble. Baseline run used --no-timesfm.
taa_no_regime	1.0993	1.0000	5.0000	Leave-one-out regime ablation.
taa_no_trend	1.1451	1.0000	5.0000	Leave-one-out trend ablation.
taa_no_momo	1.1221	1.0000	5.0000	Leave-one-out ADM ablation.
saa_minimum_variance	0.5622	nan	nan	SAA method comparison row for minimum_variance.
saa_mean_variance	0.5087	nan	nan	SAA method comparison row for mean_variance.
saa_maximum_diversification	0.5050	nan	nan	SAA method comparison row for maximum_diversification.
saa_inverse_vol	0.4897	nan	nan	SAA method comparison row for inverse_vol.
saa_risk_parity	0.4737	nan	nan	SAA method comparison row for risk_parity.
taa_baseline	2.5857	1.0000	3.0000	Baseline signal ensemble.